

Week 1 - Day 3c Supplementary: Functions

This section contains extra material related to functions

Exercise 1: Even Numbers []

1.1 Write a function `is_even` with input `x`, so that `is_even` checks if `x` is even. The function returns `True` if `x` is even, and returns `False` otherwise.

```
# TODO: Your code here
def is_even(x):
    pass
```

Run the following code cell to check your solution.

```
assert is_even(2) == True
assert is_even(-2) == True
assert is_even(3) == False

print("Test cases passed. ☑")
```

Exercise 2: Approximating Pi []

2.1 Write a function `pi` that approximates π . Use the following approximation of π :

$$\pi = 4 \left(\frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots \right)$$

Your function should get an input `n` and compute the sum of the first `n` terms in the above approximation.

Play around the input number see when it converges to π .

```
# TODO: Your code here
```

Exercise 3: Prime Numbers []

3.1 Write a function `is_prime` with input `x`. The function returns `True` if `x` is a prime, returns `False` otherwise. Please use a `for` loop instead of a `while` loop.

Hint: You can use `is_even` function you defined above.

```
# TODO: Your code here
```

Use the following code cell to check your solution.

```
assert is_prime(1) == False
assert is_prime(2) == True
assert is_prime(3) == True
assert is_prime(97) == True
assert is_prime(85) == False
assert is_prime(87) == False

print("Test cases passed. ☑")
```

Exercise 4: Prime Factors []

4.1 Write a function `factors` that, given an input `n`, returns a list contains all the prime factors of `n`. For example, for `n = 20`, your function should return `[2, 5]`. You can use the function `is_prime` from the previous question.

TODO: *Your code here*

Use the following code cell to check your solution.

```
assert sorted(factors(20)) == [2, 5]
assert sorted(factors(8)) == [2]
assert sorted(factors(1)) == []
assert sorted(factors(100)) == [2, 5]
assert sorted(factors(105)) == [3, 5, 7]

print("Test cases passed. ☑")
```

Exercise 5: Twin Primes []

Define a function `twin_primes` with no input to print all the pairs of twin primes between 2 and 200 in the form of a nested list.

For example, a pair of twin primes is both of the numbers are prime numbers and they have the difference of 2. Ex. `[5, 7]`, `[11, 13]`, `[29, 31]`

The result should be in the form of the following `[ [..., ...], [..., ...], ....]`

Use a `for` loop and the function `is_prime`.

TODO: *Your code here*

Run the following code cell to check your solution.

```
assert twin_primes() == [
    [3, 5],
    [5, 7],
    [11, 13],
    [17, 19],
```

```
[29, 31],  
[41, 43],  
[59, 61],  
[71, 73],  
[101, 103],  
[107, 109],  
[137, 139],  
[149, 151],  
[179, 181],  
[191, 193],  
[197, 199]]
```

```
print("Test cases passed. 🎉")
```

Exercise 6: Writing Functions to Generate Collections []

6.1 Write a function `spell(s)` that takes in a string `s` and returns a list of all the substrings of `s` that start from the beginning of the string. This should look like building up `s` starting from the first letter. For example:

```
spell("apple") == ["a", "ap", "app", "appl", "apple"]
```

```
# TODO: Your code here
```

6.2 Write a function `length_map(lst)` that takes in a list of strings `lst` and returns a dictionary that maps every string in the list to its length. For example:

```
length_map(["addis", "coder", "cat", "algorithm"]) == {"addis": 5,  
"coder": 5, "cat": 3, "algorithm": 9}
```

```
# TODO: Your code here
```

Exercise 7: Bubble Sort Preview []

Sorting a list is making the list goes from smallest number to the biggest one.

Bubble sort is a simple sorting procedure that repeatedly steps through the input list element by element, comparing the current element with the one after it, swapping their values if needed.

Use this algorithm to sort `lst = [9, 8, 7, 6, 5, 4, 3, 2, 1]`

Hint: There are `len(lst)` number of passes, and in each pass we need to iterate through the list to compare every adjacent pair of elements.

```
# TODO: Your code here
```

```
lst = [9, 8, 7, 6, 5, 4, 3, 2, 1]
```

Congratulations 🎉